Identifying Ignitable Liquids in Fire Debris *

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Executive Summary

This report presents a classification model designed to identify Ignitable Liquids (IL) categorized into eight classes against Substrate (SUB) backgrounds, providing statistical rigor for arson investigations. By using two key variables, the Total Ion Chromatogram (TIC) and the Total Ion Spectrum (TIS), the report found that a dual-branch CNN architecture (Model 2) significantly outperformed TIC-only baselines, increasing the macro-averaged ROC-AUC from 0.7939 to 0.8324. In the interest of integrity, the implementation of a 0.65 confidence threshold successfully prioritized the reduction of false positives by correctly identifying 90.1% of substrate-only samples in real-world testing. Furthermore, the optimal weight of zero in the hybrid model demonstrated that deep learning feature extraction is superior to a method trying to align peaks, which proved redundant. While the model achieved 98% accuracy on simulated In-Silico data, performance on real laboratory samples reached only 73%; this is a generalization gap that shows the complexity of substrate noise and the necessity of computational assistance to verify human classifications. Supplementary diagnostics and figures are provided in the Appendix.

(1) Introduction

The goal of this report is to develop a classification model capable of determining whether a fire debris sample contains an Ignitable Liquid (IL) or only Substrate (SUB). This model is intended to support crime laboratory analysts, giving statistical rigor to complement their manual classifications.

The stakes of this classification task are significant. In suspected arson cases, the detection of IL can be the determining factor in a prosecution. A false negative may allow an arsonist to evade justice, while a false positive could contribute to a wrongful conviction.

(2) Data

The two features of interest in this analysis are `TIC` and `TIS`, which will both be used to classify a given fire debris sample's `Class`. These features are extracted from three sources:

1. In-Silico Database: A simulated dataset of 60,000 samples created by computationally mixing pure IL signals with SUB signals. This serves as the training set.
2. Fire Debris (FD) Database: A real-world testing set of 1,117 samples created in a laboratory environment. These represent "ground truth" mixtures that the model must correctly identify.
3. Pure Templates: 1,275 reference signals extracted from the National Center for Forensic Science website. These represent the ideal chemical signatures for different IL classes.

Class

Samples are classified into nine possible categories. The SUB class represents a negative result (no ignitable liquid). The remaining eight classes represent specific IL groups: AR (Aromatic), GAS (Gasoline),

*Formatting based on svmille guide (http://github.com/svmiller).

PD (Petroleum Distillates), ISO (Isoparaffinic), AL (Normal Alkane), NP (Naphthenic Paraffinic), OX (Oxygenated), and MISC (Miscellaneous).

TIC

The `TIC` represents the sum of all detected ion intensities across all m/z values at each scan point.

- X-axis: Post-processed into Retention Index values, ranging from 1 to 2,800. This scale standardizes the time across different instruments.
- Y-axis: Normalized Intensity, ranging from 0.0 to 1.0. Each peak reflects when chemical components elute from the column.

TIS

The `TIS` represents the sum of ion intensities over all scan points for each mass-to-charge (m/z) ratio.

- X-axis: Mass-to-Charge Ratio (m/z), ranging from 30 to 160.
- Y-axis: Normalized Intensity, ranging from 0.0 to 1.0. For each m/z, this value is the relative number of fragments at that ratio.

(3) Exploratory Data Analysis

Distribution of Classes

Table 9 in the Appendix summarizes the distributions across the training, testing, and template datasets. The In-Silico database is perfectly balanced for binary classification (50% SUB, 50% IL). The FD database is also well-balanced for binary testing. However, the templates show a significant bias toward the PD class, which has nearly five times more representation than NP. This imbalance should be considered if any classification models are built just with the templates because training will likely be biased towards classifying samples as PD over smaller count classes like NP.

TIC Form

The `TIC` of a pure IL (such as ISO) shows distinct spikes (peaks) at specific retention indices seen in Figure 1. These peaks are the primary characteristic of the liquid that analysts consider. In both In-Silico and FD, these peaks are often disguised by the underlying noise of the SUB class (for example, this could be carpet or chair in the sample along with the IL). This makes the pattern difficult to identify manually, which is discussed later.

TIS Form

The `TIS` provides a cleaner signal for a pure IL (such as ISO again) seen in Figure 2. Figure 3 is a comparison of three randomly selected GAS samples, showing how the `TIS` patterns are highly reproducible across different samples of the same class. While analysts traditionally ignore `TIS` for visual pattern recognition, it could be incorporated as a secondary, independent confirmation of the chemical identity.

TIC Artifact

A significant artifact was identified in the `TIC` data: a continuous upward ramp at the end of the Retention Index (indices > 2000). This cluster was not able to be confirmed as being chemically significant, and should be removed because it can mislead a model into finding false patterns. An example can be seen in Figure 4.

To ensure the models only fit to legitimate chemical data, a culling algorithm was implemented. This algorithm calculates a baseline from the first 65% of the signal and identifies sustained, low-variance increases at the end of the Retention Index (low-variance to distinguish between spikes and ramps). These artifacts are zeroed out before any training or testing occurs.

Template Preprocessing

The reference CDF files containing the templates were extracted from the ILRC database. Because the raw templates use a different Retention Index baseline than the mixture datasets, they required standardizing:

1. Trimming: Low signal (leading and trailing baseline noise with intensities < 0.01) was removed.
2. Alignment: The templates were padded with zeros to reach the standard length of 2,800 points.
3. Non-Scaling: Importantly, the templates were shifted, not scaled. This preserves the exact distance between peaks, which is an important feature for IL classification.

(4) Approach One

The first approach uses a 1D Convolutional Neural Network (CNN) architecture focused exclusively on the `TIC`. This model serves as the baseline for evaluating whether the `TIC` peak patterns alone are sufficient to distinguish between IL and SUB noise.

Model 1 Explanation

The `TIC` only model takes the standardized 2,800 point Retention Index signal and passes it through a series of filters (convolutional layers). These filters are optimized to detect specific features, such as the width of a peak, the slope of a series of peaks, or the flat space between peaks.

The data used for this model consists of the normalized and cleaned `TIC` signals from the In-Silico and template databases acting as the training set, and the laboratory generated FD samples for a testing set. The final layer of the model is a Softmax classifier, which outputs a probability distribution across nine classes. For example, it might report a 70% probability of GAS and a 30% probability of SUB. To prioritize avoiding false positives, we apply a binary branching structure. We calculate the total IL Confidence ($1.0 - P(\text{SUB})$). If this value does not exceed a set threshold (for example, 0.65), the model ignores the IL classification and defaults to a result of SUB.

Model 1 Results

The results for model 1 on the real-world FD database indicate a highly conservative but reliable baseline.

The model had an overall accuracy of 68% in distinguishing IL from SUB. Crucially, it demonstrated a SUB Recall of 0.9, meaning it correctly identified 90% of the SUB samples. However, the IL Recall was lower at 0.53, suggesting that when the `TIC` signal is faint or heavily obscured by SUB noise, the model struggles to see the peaks.

Table 1: FD Database Binary Performance (TIC)

Classification	Precision	Recall	F1_Score
IL	0.889	0.526	0.661
SUB	0.558	0.901	0.69

Within the IL classes, the model performed best on GAS and ISO, which have very distinct, reproducible peak patterns. It struggled with OX and MISC samples, where patterns are less predictable.

Table 2: FD Database Multiclass Performance (TIC)

Class	Precision	Recall	F1_Score
AL	0.5	0.09	0.152
AR	0.541	0.286	0.374
GAS	0.828	0.485	0.611
ISO	0.585	0.576	0.58
MISC	0.2	0.13	0.158
NP	0.955	0.313	0.472
OX	0.09	0.109	0.098
PD	0.78	0.399	0.528
SUB	0.558	0.901	0.69

The macro-averaged ROC-AUC was 0.7939. This indicates that while the difficult classifications were conservative, the underlying probability scores were good at separating classes. A score of ≈ 0.80 suggests the model is a strong classifier.

Checking model performance by testing on the In-Silico dataset that it was trained on shows that the model likely fit the noise or simulation features to some extent, with a very high macro-averaged ROC-AUC of 0.9964 and overall accuracy of 98% in distinguishing IL from SUB.

Table 3: In-Silico Binary Performance (TIC)

Classification	Precision	Recall	F1_Score
IL	0.999	0.956	0.977
SUB	0.958	0.999	0.978

Additionally, the precision for classifying each class in In-Silico is also significantly higher than for FD database.

Table 4: In-Silico Multiclass Performance (TIC)

Class	Precision	Recall	F1_Score
AL	0.988	0.973	0.981
AR	0.96	0.939	0.949
GAS	0.978	0.969	0.973
ISO	0.96	0.962	0.961
MISC	0.874	0.763	0.815

Class	Precision	Recall	F1_Score
NP	0.944	0.945	0.944
OX	0.934	0.81	0.868
PD	0.917	0.88	0.898
SUB	0.958	0.999	0.978

(5) Approach Two

The second approach changes the model into a Multi-Input architecture that considers both the TIC and the TIS.

Model 2 Explanation

This model uses a branching architecture. Instead of a single processing path, the model splits into two parallel branches for TIC and TIS data. Importantly, there is no fixed mathematical mapping between Retention Index and specific m/z values. By keeping the branches separate until the final fusion layer, the model learns patterns in the temporal structure (TIC) and fragmentation structure (TIS) independently. This allows the model to remain robust to shifts in Retention Index caused by weathering, while still recognizing the mass spectrum signals. After each branch identifies its own unique patterns, the model joins these summaries together so a final layer can weigh all the evidence and decide which class is the most likely match.

Model 2 Results

The addition of TIS data provided a significant boost to the model's performance, particularly in its ability to detect low signal ignitable liquids.

The binary accuracy increased to 73% on the FD database. While the SUB Recall remained strong at 0.80, the IL Recall rose significantly to 0.68 (up from 0.53 in Approach One). This shows that the TIS provides additional information that allows the model to identify an IL even when the TIC peaks are buried under substrate noise.

Table 5: FD Database Binary Performance (TIC+TIS)

Classification	Precision	Recall	F1_Score	Accuracy
IL	0.837	0.681	0.751	0.729
SUB	0.625	0.8	0.702	0.729

The multiclass F1-scores improved across almost every category (F1-score $= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$, where high scores indicate a good balance between precision and recall). For example, NP recall rose to 0.597, nearly doubling the performance of the TIC -only approach, while its precision only decreased by 0.085 from the previous model. This is likely because many IL classes have similar TIC shapes but very different consistent TIS shapes.

Table 6: FD Database Multiclass Performance (TIC+TIS)

Class	Precision	Recall	F1_Score
AL	0.453	0.507	0.479
AR	0.725	0.414	0.527
GAS	0.819	0.596	0.69
ISO	0.576	0.515	0.544
MISC	0.319	0.333	0.326
NP	0.87	0.597	0.708
OX	0.033	0.073	0.045
PD	0.902	0.309	0.46
SUB	0.625	0.8	0.702

The macro-averaged ROC-AUC rose to 0.8324. This increase indicates that the dual-input approach is more robust. The model is now 83% likely to rank a true IL sample higher than a SUB sample.

The model performance from testing on the In-Silico dataset is basically the same as before, with a macro-averaged ROC-AUC of 0.9948 and overall accuracy of 98% in distinguishing IL from SUB.

Table 7: In-Silico Binary Performance (TIC+TIS)

Classification	Precision	Recall	F1_Score	Accuracy
IL	0.993	0.971	0.982	0.982
SUB	0.972	0.993	0.982	0.982

No noticeable differences exist for the multiclass results tested on In-Silico either.

Table 8: In-Silico Multiclass Performance (TIC+TIS)

Class	Precision	Recall	F1_Score
AL	0.934	0.99	0.961
AR	0.968	0.914	0.94
GAS	0.969	0.969	0.969
ISO	0.964	0.953	0.959
MISC	0.755	0.723	0.739
NP	0.947	0.945	0.946
OX	0.828	0.89	0.858
PD	0.925	0.739	0.822
SUB	0.972	0.993	0.982

(6) Approach Three

The final approach is a hybrid model that uses the result of the Multi-Input CNN model with a new Alignment algorithm. This approach acts as a final layer, where reference templates from the ILRC database provide additional evidence to refine the final classification of a sample identified as IL by Model 2.

Model 3 Explanation

The hybrid model operates in two distinct phases: extraction (1) and scoring (2).

1. For every sample and reference template, we extract the most significant peaks. For TIC, 12 points are identified with a required minimum distance of 20 units. For TIS, 8 points are identified with a minimum distance of 2 units. Each peak is assigned a weight proportional to its height ($height^{2.5}$), which makes sure that actual peaks have more significance than minor fluctuations.

2. The sample is slid against every template in each class to find the best alignment. This is done independently for TIC and TIS, with the resulting scores summed to represent the total evidence for a specific IL class. To calculate the alignment at each step, a multivariate Gaussian distribution is applied around every peak, defining a region of probable overlap in both time (or m/z) and intensity. The final score is calculated by multiplying these individual point distributions, effectively rewarding samples where multiple peaks align closely with the reference template. To prevent classes with many templates (like PD) from dominating, we use the average score across only the strongest 10% of matches for each class.

The final classification is determined by a weight (α). The model combines the CNN's probability distribution with the alignment scores:

$$\text{Combined Score} = P(\text{CNN}) + (\alpha \times \text{Peak Evidence})$$

The hyperparameter α represents how much additional trust we give to the direct template matching over the neural network's internal patterns.

Model 3 Results

The training process for the hybrid model involves finding the value of α that maximizes classification accuracy on the In-Silico dataset for samples already identified as IL (the range of α values considered was 0 to 2 by 0.1).

The optimization process yielded an optimal alpha of 0. This indicates that, for the current dataset, the Multi-Input CNN has already achieved such a high level of feature extraction that adding direct template alignment scores does not provide additional power.

Because the CNN was trained on the In-Silico data (which was generated using these reference templates) the network has already internalized the template peaks. While the alignment scores provide an interesting secondary validation that could be considered, they are redundant for increasing the overall accuracy beyond the 90.25% already achieved by the neural network for samples already identified as IL by the model.

(7) Model Comparisons

While Model 1 provided a strong baseline using only TIC data, Model 2 emerged as the superior estimator. Model 2 outperformed the baseline with a macro-averaged ROC-AUC of 0.8324, compared to Model 1's 0.7939. This improvement suggests that integrating TIS enables the network to resolve cases that were previously visually ambiguous. Approach Three, the hybrid model, ultimately reverted to the weights of Model 2, confirming that it does not improve upon Model 2.

Tradeoffs and Error Costs

In the context of arson investigation, the cost of misclassification should be discussed. Classifying a sample as containing an IL when it is actually just SUB (a Type I error) could lead to a false conviction. This outcome is ethically and legally far more damaging than a Type II error (failing to detect an IL that is present).

While Model 2 showed superior sensitivity, its SUB precision of 0.625 on the FD database implies that about 37% of samples predicted as SUB are actually IL, and its SUB recall of 0.800 indicates that about 20% of true SUB samples are misclassified as IL. This highlights the importance of the binary boundary used. Currently, the threshold is set at 0.65; however, in a real-world application, this boundary should be treated as a tunable parameter. By increasing this threshold, we can intentionally force the model to be more skeptical, requiring more evidence before allowing a sample to be classified as an IL. This allows the analysts to define a specific, known error rate that prioritizes the protection of the innocent, even at the cost of overall accuracy.

Skepticism of Visual Analysis

The results of Approach Three provide a compelling reason to remain skeptical of traditional manual analysis. The alignment algorithm, which mimics the human process of identifying and matching major peaks to known reference standards, proved to be redundant when compared to the patterns identified by the CNN. The fact that an α of 0 was optimal suggests that simple peak-matching is an insufficient solution for the complex patterns present in fire debris.

Furthermore, while the CNNs represent a significant leap over manual template matching, their performance on real-world laboratory samples remains disappointing. An accuracy of $\approx 73\%$ in a binary setting is a powerful tool for investigative leads, but it falls short of the "beyond a reasonable doubt" standard required in court. If a highly optimized neural network trained on 60,000 samples struggles to resolve these signals perfectly, it brings into serious question the reliability of human analysts performing visual pattern recognition without assistance.

(8) Conclusions

I have divided the conclusion into a recap of the results from the analysis and a second section that presents recommendations for any future iterations of this study.

Results

The transition from a single-feature baseline (Model 1) to a dual-branch architecture (Model 2) provided the most significant performance gain. By considering both the `TIC` and the `TIS` independently, the model achieved a macro-averaged ROC-AUC of 0.8324, a substantial increase over the 0.7939 achieved by `TIC` data alone. This shows that previously unused `TIS` patterns provide additional information that can resolve `TIC` ambiguities.

The implementation of a binary branching structure with a confidence threshold of 0.65 proved successful in maintaining forensic integrity. Model 2 correctly identified 80% of SUB samples in real-world laboratory data, ensuring that the model remains skeptical of noisy backgrounds that lack a clear signal.

Approach Three revealed that the neural network's internalized feature extraction is more robust than a direct alignment algorithm. The finding of an optimal α of 0 demonstrates that deep learning can capture complex, non-linear relationships in weathered and masked data that traditional peak matching (the method closest to human visual analysis) cannot effectively replicate.

While the models achieved near-perfect accuracy (up to 98%) on simulated mixtures they were trained on, performance leveled off at approximately 73% binary accuracy when tested on laboratory-generated fire debris. This generalization gap shows the complexity of SUB interference in actual fire scenes.

Recommendations

Given the skepticism I have expressed regarding manual analysis in this report, I recommend that a performance evaluation be conducted using a panel of certified fire debris analysts. By having experts classify the same In-Silico and FD database samples, we can establish a reliable human baseline. This comparison would quantify the performance differences between traditional visual analysis and machine learning models, providing important context to courtrooms.

To decrease the risk of misidentifying the sample because of site-specific noise, investigators should implement a multi-sample collection strategy based on the fire's point of origin, which can often be determined by investigators.[1] By taking samples at increasing distances from the suspected start point, the model could be trained to recognize a gradient; identifying a consistent SUB pattern where the IL signal is strongest at the origin and progressively weakens further away. This spatial context would hopefully make the model more robust against local chemical interference.

Appendix

EDA Figures

Table 9: Sample Counts by Class

Class	In-Silico	FD	Templates
SUB	30000	446	0
AR	3750	70	94
GAS	3750	99	116
PD	3750	178	338
ISO	3750	66	106
AL	3750	67	79
NP	3750	67	70
OX	3750	55	204
MISC	3750	69	211
Total	60000	1117	1218

[1] Cunningham (2026) video.

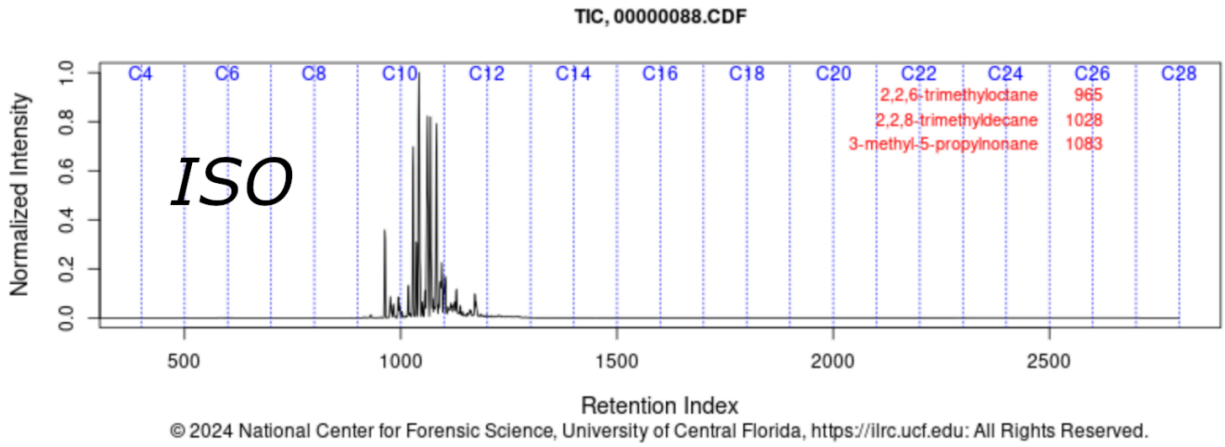

Figure 1: TIC Example

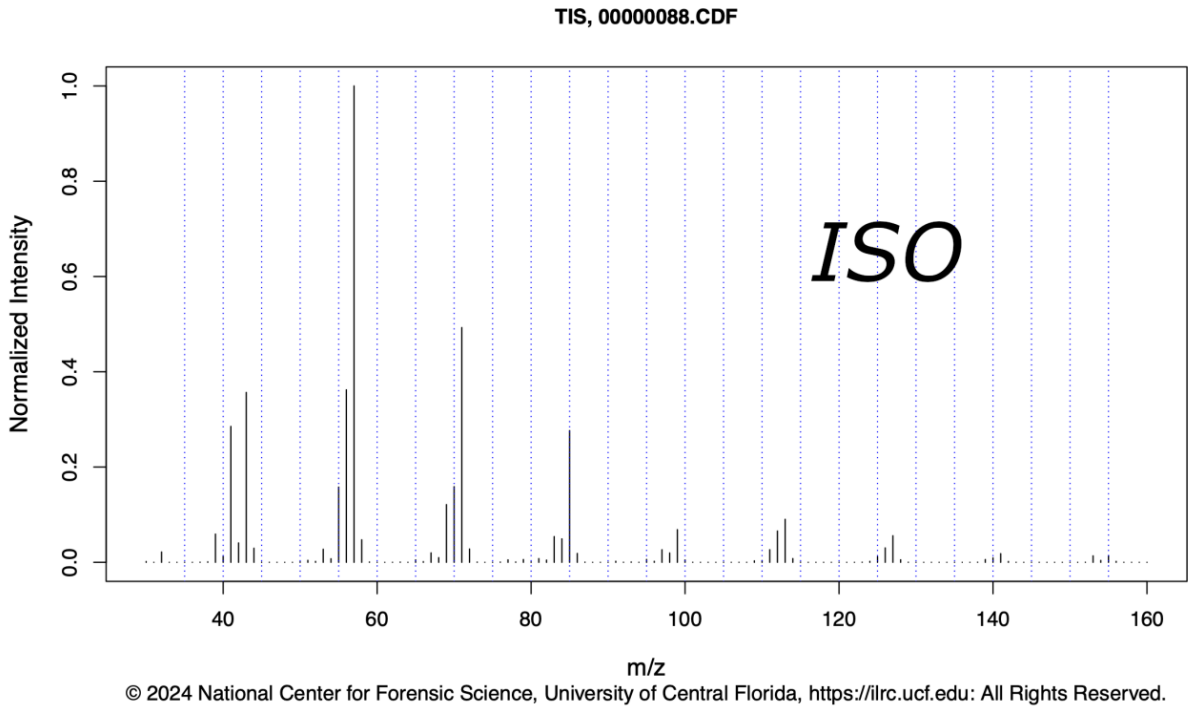

Figure 2: TIS Example

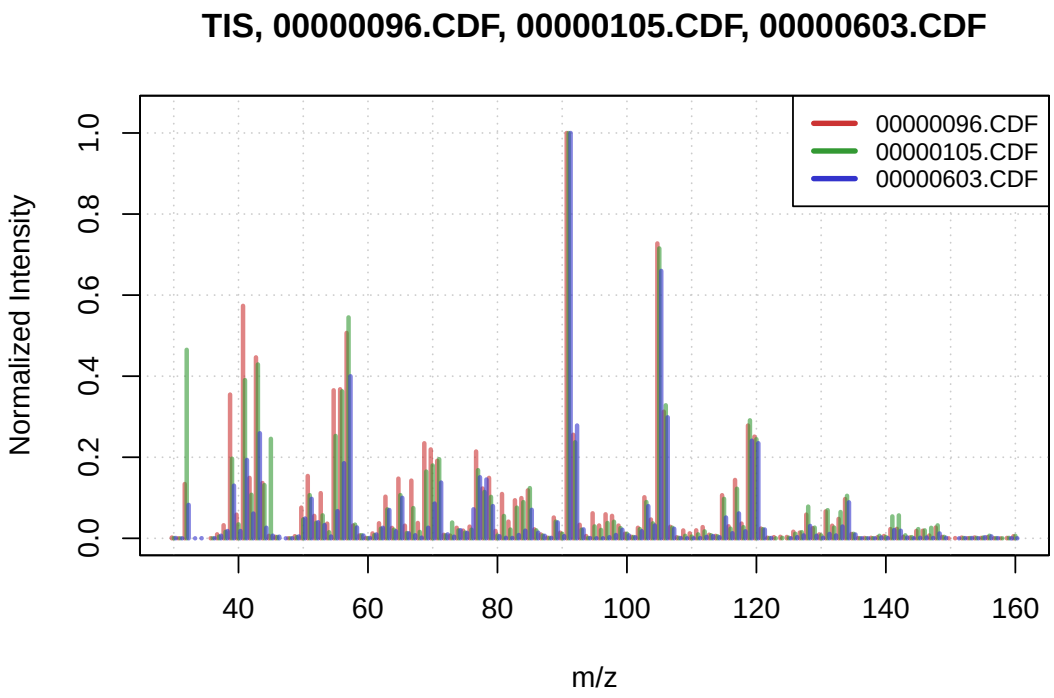

Figure 3: TIS Comparison

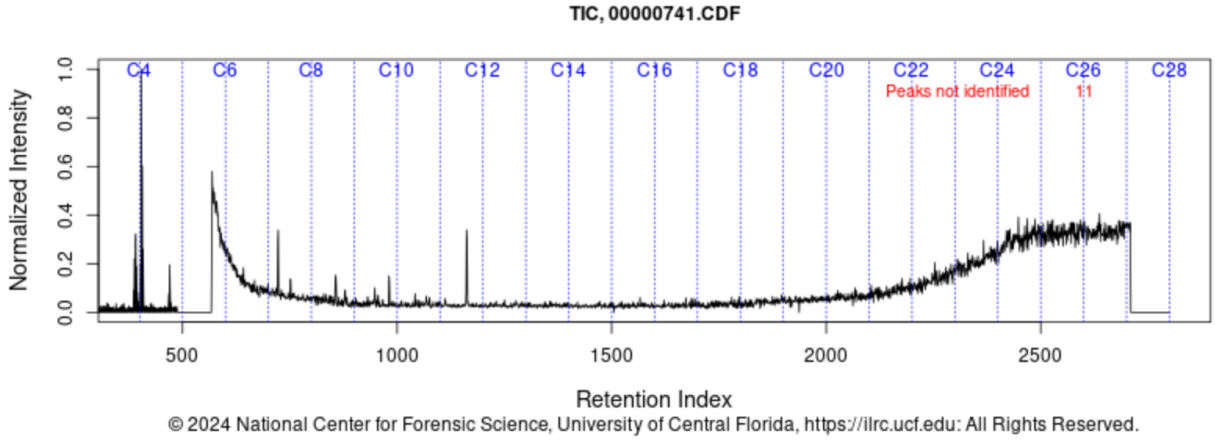

Figure 4: TIC Example

References

Cunningham, Ewan. 2026. “How Investigators Find the Source of a Fire.”.